UCSD ECE 268 - Security of Hardware Embedded Systems

Merkle Tree with Post-Quantum Secure Hash

KangarooTwelve and Rescue-Prime

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Background & Motivation

Merkle Tree:

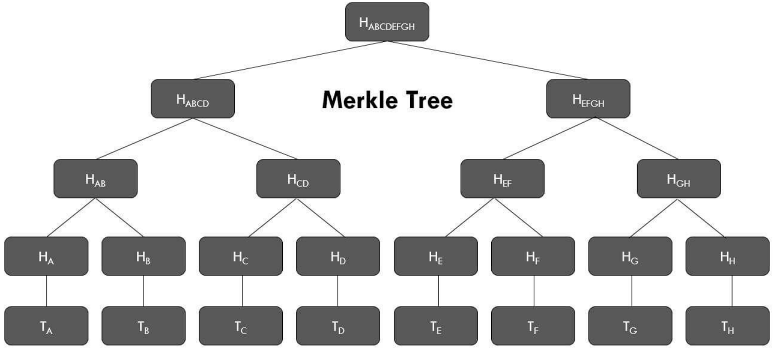
- Hierarchical cryptographic data structure
- Leaves = hash of data
- Parents = hash of children

Kangaroo12:

- Need for modern standard with high throughput on standard hardware
- Based on SHA3 permutation,utilizes relies heavily on operations

Rescue-Prime:

- Meant for zero-knowledge proving systems
- Operates over large prime fields relying on modular arithmetic and matrix operations

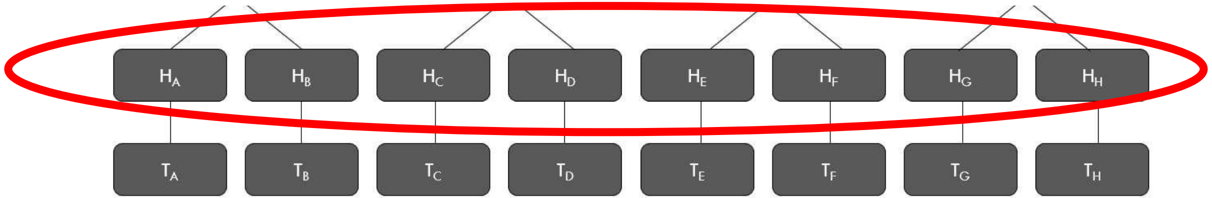


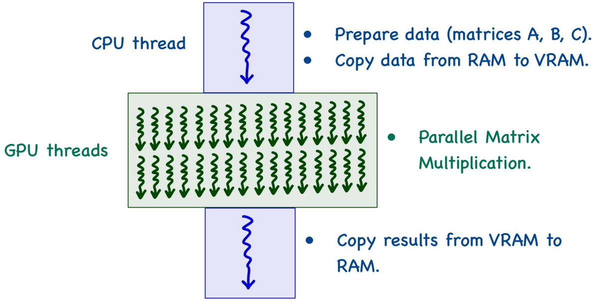
Approach

Merkle Tree: exploit parallelism on GPU by level-batched hashing

KangarooTwelve: parallelize Keccak permutations for chaining values, exploit any optimizations for bitwise operations

Rescue-Prime: can exploit parallelism of matrix multiplication in the MDS Matrix Multiplication layer of the Rescue-XLIX permutations and operations mapped accross a state





Results, Observations, Next Steps

Sequential hashing data blocks (eg. a single level on Merkle Tree) with GPU-optimized hash functions did not necessarily yield speedup.

```
--- CPU Performance Results ---
Total time (5 passes): 1.455944 seconds
Average time per hash     : 0.291189 seconds

--- GPU Performance Results ---
Total time (5 passes): 29.735436 seconds
Average time per hash     : 5.947087 seconds
```

Example with Rescue-Prime Yielding 0.05x speedup (GPU vs CPU)

Batching across a level yielded drastically improved performance.

```
Running CPU Batch...
[-] CPU Batch completed in: 39.3482 seconds

Running GPU Batch...
[-] GPU Batch completed in: 0.6550 seconds
```

Example with Rescue-Prime Yielding ~60x speedup! (GPU vs CPU)

KangarooTwelve underperforms significantly on GPU, even when batched.

Next steps: continue work on GPU optimizations on KangarooTwelve

```
Processing workload containing 10000 messages...
[-] CPU Processing batch took: 1.0191 seconds
[-] GPU Processing batch took: 40.0672 seconds
```

Example with KangarooTwelve Yielding ~0.03x speedup (GPU vs CPU)